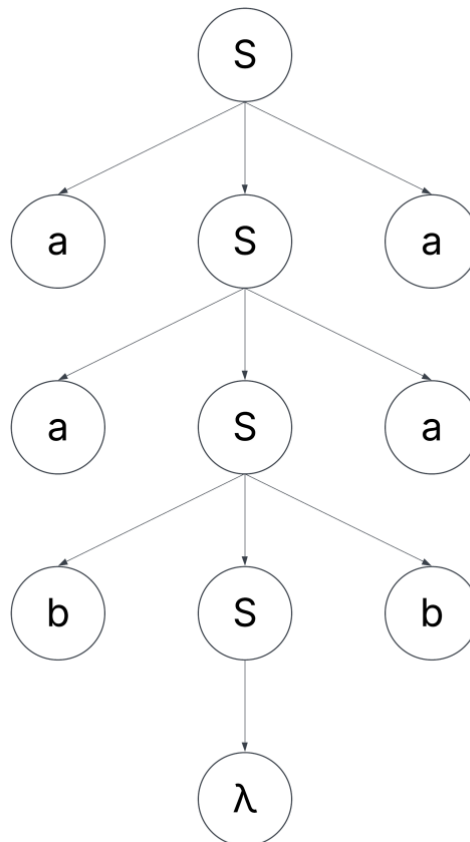


MET CS662 - Assignment #6

1. Draw the derivation tree corresponding to the derivation in Example 5.1, that is

$$S \Rightarrow aSa \Rightarrow aaSaa \Rightarrow aabSbaa \Rightarrow aabbbaa$$

where the grammar is $S \rightarrow aSa \quad S \rightarrow bSb \quad S \rightarrow \lambda$



2. Find context-free grammars for the language $L = \{a^n b^m c^k \mid n = m \text{ or } m \leq k\}$, (with also, $n \geq 0, m \geq 0, k \geq 0$)

$$\begin{aligned} S &\rightarrow AB|CD \\ A &\rightarrow aAb|\lambda \\ B &\rightarrow Bc|\lambda \\ C &\rightarrow aC|\lambda \\ D &\rightarrow bDc|E \\ E &\rightarrow Ec|c \end{aligned}$$

3. Let $L = \{a^n b^n \mid n \geq 0\}$.

a) Show that L^2 is context-free.

$$\begin{aligned} S &\rightarrow S_1 S_1 \\ S_1 &\rightarrow a S b \mid \lambda \end{aligned}$$

b) Show that L^k is context-free for any given $k \geq 1$.

$$\begin{aligned} S &\rightarrow S_1 S_1 S_1 \dots S_1 S_1 S_1; k \text{ times} \\ S_1 &\rightarrow a S b \mid \lambda \end{aligned}$$

c) Show that $\bar{L}$ and L^* are context-free.

$$\begin{aligned} \bar{L} &= \{awb \mid w \notin L\} \cup \{bw \mid w \in \{a, b\}^*\} \cup \{wa \mid w \in \{a, b\}^*\} \\ S &\rightarrow a S b \mid b A \mid A a \\ A &\rightarrow a A \mid b A \mid \lambda \end{aligned}$$

$$\begin{aligned} L^* &= \{a^n b^n a^m b^m a^l b^l \dots a^k b^k \mid n \geq 0, m \geq 0, l \geq 0, \dots, k \geq 0\} \\ S &\rightarrow S a S b S \mid \lambda \end{aligned}$$

4. Find an s-grammar for the language $L(aaa^*b + b)$.

$$\begin{aligned} S &\rightarrow aAB \mid b \\ A &\rightarrow a \\ B &\rightarrow aB \mid b \end{aligned}$$

5. Show that the following language is ambiguous.

$$S \rightarrow AB \mid aaB \quad A \rightarrow a \mid Aa \quad B \rightarrow b$$

$$\begin{aligned} S &\rightarrow aaB \rightarrow aab \\ S &\rightarrow AB \rightarrow AaB \rightarrow aaB \rightarrow aab \end{aligned}$$

6. Construct an unambiguous grammar equivalent to the grammar

$$S \rightarrow AB \mid aaB \quad A \rightarrow a \mid Aa \quad B \rightarrow b$$

$$\begin{aligned} S &\rightarrow AB \\ A &\rightarrow a \mid Aa \\ B &\rightarrow b \end{aligned}$$

7. Prove the following result. Let $G = (V, T, S, P)$ be a context-free grammar in which every $A \in V$ occurs on the left side of at most one production. Then G is unambiguous.

If every $A \in V$ occurs on the left side of at most one production, then there is no choice when parsing the grammar. If there is no choice when parsing the grammar, then there is only one possible derivation tree. If there is only one possible derivation tree, then the grammar is unambiguous.

8. Transform the grammar with productions $S \rightarrow abAB$, $A \rightarrow bAB|\lambda$, $B \rightarrow BAa|A|\lambda$

into Chomsky normal form.

$$\begin{aligned} S &\rightarrow CE \\ A &\rightarrow DF|b \\ B &\rightarrow AC|BG|DF|a|b \\ C &\rightarrow a \\ D &\rightarrow b \\ E &\rightarrow DF|b \\ F &\rightarrow AB|AC|BG|DF|a|b \\ G &\rightarrow AC|a \end{aligned}$$

9. Convert the grammar $S \rightarrow aSb|bSa|a|b$

into Greibach form.

$$\begin{aligned} S &\rightarrow aSB|bSA|a|b \\ A &\rightarrow a \\ B &\rightarrow b \end{aligned}$$
